Sports Tournament Ranking System

Python Programming – Activity-Based Learning (ABL) Assignment

1. Problem Statement

Store sports match results and team scores, calculate total points and tie-break values, and produce rankings according to a specified competition policy.

2. Objectives

• Store each match result.

• Calculate matches played, wins, draws, losses, goals for, goals against, goal difference, and points.

• Sort teams using the competition tie-break rules.

3. Assumptions and Competition Policy

A win earns 3 points, a draw earns 1 point for each team, and a loss earns 0 points. Teams are ranked by (1) total points, (2) goal difference (GF - GA), (3) goals scored (GF), and (4) team name alphabetically. The sample tournament contains six matches and four teams.

4. Algorithm

1. Create an empty dictionary to store each team's statistics.

2. Read each match and create team records if they do not exist.

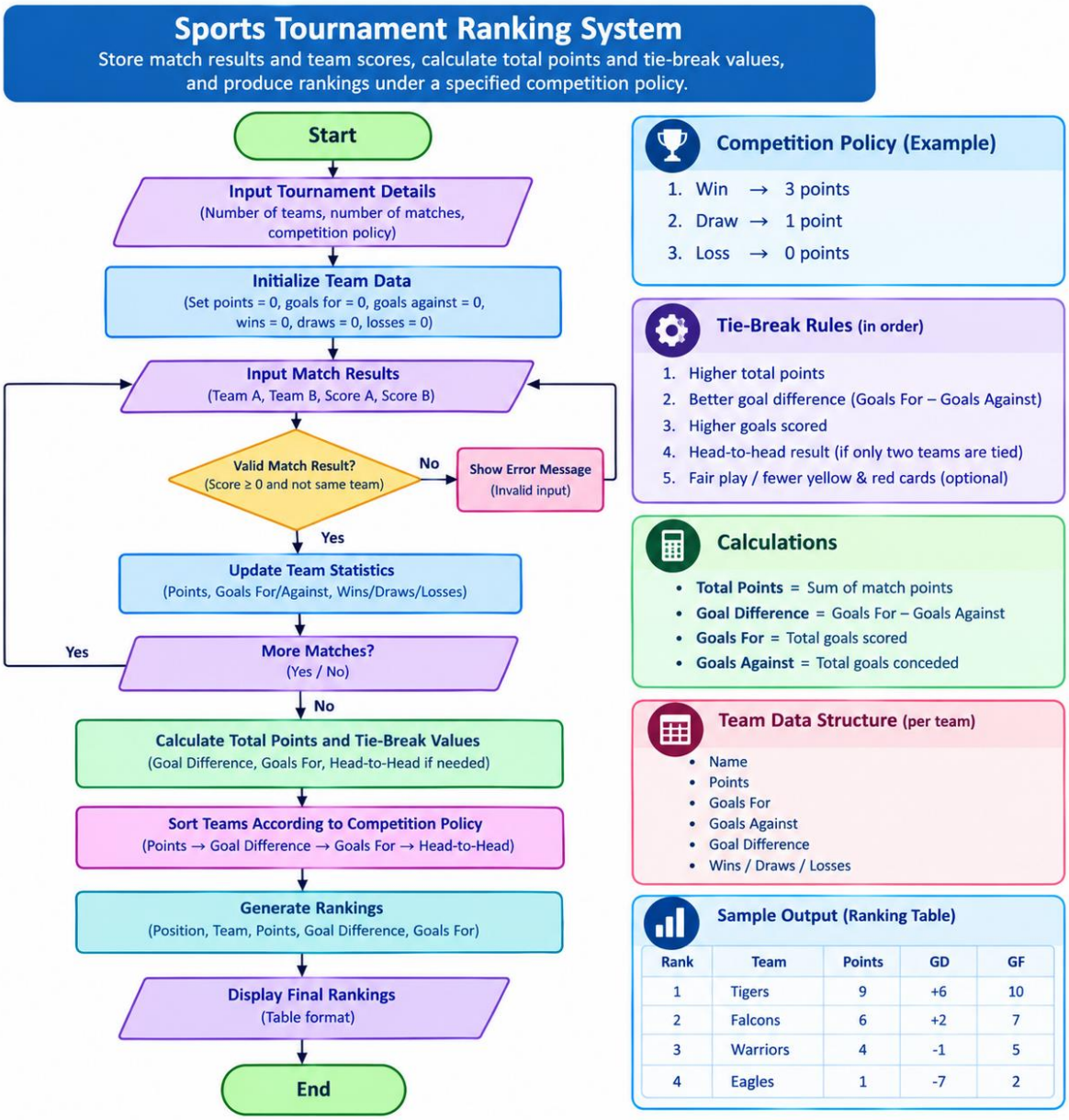
3. Update played matches, goals for, and goals against.

4. Award points and update win/draw/loss counts based on the score.

5. Sort teams by points descending, goal difference descending, goals scored descending, and team name ascending.

6. Display the ranked table and a completion message.

5. Flowchart



6. Pseudocode

BEGIN

INPUT number_of_teams

```
CREATE empty list of teams

FOR each team from 1 to number_of_teams DO
   INPUT team_name

   SET points = 0
   SET goals_for = 0
   SET goals_against = 0
   SET wins = 0
   SET draws = 0
   SET losses = 0

   STORE team details
END FOR

INPUT number_of_matches

FOR each match from 1 to number_of_matches DO

   INPUT team_A
   INPUT team_B
   INPUT score_A
   INPUT score_B

   IF team_A = team_B OR
     score_A < 0 OR score_B < 0 THEN
```

```
    DISPLAY "Invalid Match Result"
    CONTINUE

END IF

// Update Goals
team_A.goals_for += score_A
team_A.goals_against += score_B

team_B.goals_for += score_B
team_B.goals_against += score_A

// Award Points
IF score_A > score_B THEN

    team_A.points += 3
    team_A.wins += 1
    team_B.losses += 1

ELSE IF score_A < score_B THEN

    team_B.points += 3
    team_B.wins += 1
    team_A.losses += 1
```

```
    ELSE

        team_A.points += 1
        team_B.points += 1

        team_A.draws += 1
        team_B.draws += 1

    END IF

    STORE match result

END FOR

// Calculate Tie-Break Values
FOR each team DO
    team.goal_difference =
        team.goals_for - team.goals_against
END FOR

// Rank Teams According to Competition Policy
SORT teams BY
    points DESCENDING,
    goal_difference DESCENDING,
    goals_for DESCENDING
```

```
    // Apply Head-to-Head Rule if Required
    RESOLVE remaining ties using competition policy

    // Display Final Rankings
    DISPLAY "SPORTS TOURNAMENT RANKINGS"

    rank = 1

    FOR each team IN sorted teams DO
        DISPLAY rank
        DISPLAY team.name
        DISPLAY team.points
        DISPLAY team.goal_difference
        DISPLAY team.goals_for
        DISPLAY team.wins
        DISPLAY team.draws
        DISPLAY team.losses

        rank = rank + 1
    END FOR
    DISPLAY "Tournament Ranking Completed"

END
```

7. Python Source Code

```
def create_team(name):
```

```
return {
        "name": name,
        "points": 0,
        "goals_for": 0,
        "goals_against": 0,
       "goal_difference": 0,
        "wins": 0,
        "draws": 0,
         "losses": 0
}
def record_match(teams, match_results):
 team_a = input("Enter Team A: ").strip()
team_b = input("Enter Team B: ").strip()
    print("Error: Team not found!")
    return

if team_a == team_b:
    print("Error: A team cannot play against itself!")
    return

try:
    score_a = int(input(f"Enter {team_a} score: "))
    score_b = int(input(f"Enter {team_b} score: "))

    if score_a < 0 or score_b < 0:
        print("Error: Scores cannot be negative!")
        return

except ValueError:
```

```
    print("Error: Enter valid whole-number scores!")
    return

a = teams[team_a]
b = teams[team_b]

# Update goals
a["goals_for"] += score_a
a["goals_against"] += score_b

b["goals_for"] += score_b
b["goals_against"] += score_a

# Award points
if score_a > score_b:
    a["points"] += 3
    a["wins"] += 1
    b["losses"] += 1

elif score_b > score_a:
    b["points"] += 3
    b["wins"] += 1
    a["losses"] += 1

else:
    a["points"] += 1
    b["points"] += 1
    a["draws"] += 1
    b["draws"] += 1

# Update goal difference
a["goal_difference"] = (
    a["goals_for"] - a["goals_against"]
)
b["goal_difference"] = (
    b["goals_for"] - b["goals_against"]
)

match_results.append({
```

```
    "team_a": team_a,
    "team_b": team_b,
    "score_a": score_a,
    "score_b": score_b
})

print(f"Match recorded: {team_a} {score_a} - "
      f"{score_b} {team_b}")
```

def display_match_results(match_results): print("\n========== MATCH RESULTS ==========")

```
if not match_results:
    print("No matches recorded.")
    return

for i, match in enumerate(match_results, start=1):
    print(
        f"{i}. {match['team_a']} "
        f"{match['score_a']} - {match['score_b']} "
        f"{match['team_b']}"
    )
```

def display_rankings(teams):

Ranking policy: # 1. Total points (descending) # 2. Goal difference (descending) # 3. Goals scored (descending) ranking = sorted(teams.values(), key=lambda t: (t["points"], t["goal_difference"], t["goals_for"]), reverse=True)

```
print("\n" + "=" * 100)
print("                  FINAL TOURNAMENT RANKINGS")
print("=" * 100)

print(
    f"{'Rank':<6}{'Team':<15}{'P':>4}"
    f"{'W':>4}{'D':>4}{'L':>4}"
    f"{'GF':>7}{'GA':>7}{'GD':>7}{'Points':>9}"
)
```

```
print("-" * 100)

for rank, team in enumerate(ranking, start=1):
    print(
        f"{rank:<6}{team['name']:<15}"
        f"{team['wins']:>4}{team['wins'] + team['draws'] +
team['losses']:>4}"
        f"{team['draws']:>4}{team['losses']:>4}"
        f"{team['goals_for']:>7}{team['goals_against']:>7}"
        f"{team['goal_difference']:>7}{team['points']:>9}"
    )

print("=" * 100)


def main():

     print("=" * 60)

     print(" SPORTS TOURNAMENT RANKING SYSTEM")

     print("=" * 60)

teams = {}
match_results = []

try:
    n = int(input("Enter number of teams: "))
    if n < 2:
        print("At least 2 teams are required.")
        return
except ValueError:
    print("Enter a valid number.")
    return

# Register teams
for i in range(n):
    name = input(f"Enter name of Team {i + 1}: ").strip()

    if not name or name in teams:
```

```
            print("Invalid or duplicate team name. Try again.")
            return

        teams[name] = create_team(name)

    while True:
        print("\n---------- MENU ----------")
        print("1. Record Match Result")
        print("2. View Match Results")
        print("3. View Team Rankings")
        print("4. Exit")

        choice = input("Enter choice (1-4): ").strip()

        if choice == "1":
            record_match(teams, match_results)

        elif choice == "2":
            display_match_results(match_results)

        elif choice == "3":
            display_rankings(teams)

        elif choice == "4":
            display_rankings(teams)
            print("Tournament ranking completed.")
            break

        else:
            print("Invalid choice. Select 1, 2, 3, or 4.")


if __name__ == "__main__": main()
```

8. Output Screenshot

```
============================================================
       SPORTS TOURNAMENT RANKING SYSTEM
============================================================
Enter number of teams: 4
Enter name of Team 1: Tigers
Enter name of Team 2: Falcons
Enter name of Team 3: Warriors
Enter name of Team 4: Eagles

---------- MENU ----------
1. Record Match Result
2. View Match Results
3. View Team Rankings
4. Exit
Enter choice (1-4): 1
Enter Team A: Tigers
Enter Team B: Falcons
Enter Tigers score: 3
Enter Falcons score: 1
Match recorded: Tigers 3 - 1 Falcons

---------- MENU ----------
1. Record Match Result
2. View Match Results
3. View Team Rankings
4. Exit
Enter choice (1-4): 1
Enter Team A: Warriors
Enter Team B: Eagles
Enter Warriors score: 2
Enter Eagles score: 2
Match recorded: Warriors 2 - 2 Eagles
```

```
---------- MENU ----------
1. Record Match Result
2. View Match Results
3. View Team Rankings
4. Exit
Enter choice (1-4): 1
Enter Team A: Tigers
Enter Team B: Warriors
Enter Tigers score: 1
Enter Warriors score: 0
Match recorded: Tigers 1 - 0 Warriors

---------- MENU ----------
1. Record Match Result
2. View Match Results
3. View Team Rankings
4. Exit
Enter choice (1-4): 3

================================================================================
============
                FINAL TOURNAMENT RANKINGS
================================================================================
============
Rank  Team              P   W   D   L     GF     GA     GD   Points
--------------------------------------------------------------------------------
------------
1     Tigers            2   2   0   0      4      1      3        6
2     Eagles            0   1   1   0      2      2      0        1
3     Warriors          0   2   1   1      2      3     -1        1
4     Falcons           0   1   0   1      1      3     -2        0
================================================================================
============
```

8. Test Cases

1. **Team Registration:** Verify that new teams can be registered successfully in the tournament system.

2. **Record Match Results:** Check whether the system correctly stores match scores for two teams.

3. **Points Calculation for Win:** Verify that the winning team receives 3 points and the losing team receives 0 points.

4. **Points Calculation for Draw:** Check whether both teams receive 1 point each when a match ends in a draw.

5. **Goal Difference Calculation:** Verify that goal difference is calculated correctly using Goals For (GF) – Goals Against (GA).

6. **Ranking by Total Points:** Check whether teams are ranked in descending order of total points.

7. **Tie-Breaker Using Goal Difference:** Verify that when teams have equal points, the team with the higher goal difference is ranked higher.

8. **Tie-Breaker Using Goals Scored:** Check whether the team that scores more goals is ranked higher when both points and goal difference are equal.

9. **Display Team Statistics:** Verify that the system displays matches played, wins, draws, losses, goals scored, goals conceded, goal difference, and total points correctly.

10. **Invalid Match Scores:** Check whether the system rejects invalid scores, such as negative goals or non-numeric input.

11. **Unknown Team Validation:** Verify that match results cannot be recorded for teams that have not been registered.

12. **Multiple Match Results:** Check whether points, scores, and rankings are updated correctly after multiple matches.

13. **Ranking with Equal Statistics:** Verify that the system handles teams with equal points, goal difference, and goals scored according to the specified competition policy.

14. **No Match Results:** Check whether the system displays appropriate rankings when teams are registered but no matches have been played.

15. **Exit Option:** Verify that the system terminates correctly when the user selects the exit option.

10. Conclusion

The program successfully processes match results, calculates team statistics and points, applies the specified tie-break rules, and prints a final tournament ranking. The scoring policy can be changed by updating the constants and sorting key.